DLD Homework 1 Solution

Q1)

a) $(1101.011)_2$ =

$(1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) + (0 \times 2^{-1}) + (1 \times 2^{-2}) + (1 \times 2^{-3}) = (13.375)_{10}$

b) $(ABC.D3)_{15}$ =

$(10 \times 15^2) + (11 \times 15^1) + (12 \times 15^0) + (13 \times 15^{-1}) + (3 \times 15^{-2}) = (2427.88)_{10}$

c) $(123.01)_4$ =

$(1 \times 4^2) + (2 \times 4^1) + (3 \times 4^0) + (0 \times 4^{-1}) + (1 \times 4^{-2}) = (27.0625)_{10}$

d) $(62.73)_{20}$ =

$(6 \times 20^1) + (2 \times 20^0) + (7 \times 20^{-1}) + (3 \times 20^{-2}) = (122.3575)_{10}$

Q2)

a)

Whole number Part:

Division	Quotient	Remainder
33/2	16	1
16/2	8	0
8/2	4	0
4/2	2	0
2/2	1	0
1/2	0	1

$(0.3)_{10}$ to binary:

Multiplication	Answer	First Number
0.3x2	**0**.6	0
0.6x2	**1**.2	1
0.2x2	**0**.4	0
0.4x2	**0**.8	0
0.8x2	**1**.6	1

0.6x2	**1**.2	1

$(100001.010011)_2$

b)

$(3.21)_4 = (3 \times 4^0) + (2 \times 4^{-1}) + (1 \times 4^{-2}) = (3.5625)_{10}$

$(3)_{10}$ to octal is $(3)_8$ (Whole number part)

0.5625 (Decimal Part Conversion

Multiplication	Answer	First Number
0.5625x8	**4**.5	4
0.5x8	**4**.0	4

$(3.21)_4 = (3.44)_8$

c) $(1.333)_8 = (1 \times 8^0) + (3 \times 8^{-1}) + (3 \times 8^{-2}) + (3 \times 8^{-3}) = (1.427734375)$ Part: $(1)_{10} = (1)_{16}$

0.427734375

Multiplication	Answer	First Number
0.427734375x16	**6**.84375	6
0.84375x16	**13**.5	13 (D)
0.5x16	**8**	8

$(1.6D8)_{16}$

d) $(0101.100)_2 = (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) + (1 \times 2^{-1}) + (0 \times 2^{-2}) + (0 \times 2^{-3}) = (5.5)_{10}$

$(5)_{10} = (5)_9$

0.5:

Multiplication	Answer	First Number
0.5x9	**4**.5	4
0.5x9	**4**.5	4
0.5x9	**4**.5	4
0.5x9	**4**.5	4

$(5.4444)_9$

Q3)

a) $(1 \times 2^7) + (0 \times 2^6) + (1 \times 2^5) + (0 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) = 165$
b) Signed number (2's complement) = 01011011 = -91
c) Signed number (1's complement) = 01011010 = -90
d) Invalid, 1010 is not a valid BCD number

Q4)

a)

Actual	0	1	2	3	4	5	6	7	8	9	10	11	...
Coded	%	$	#	@	!	$%	$$	$#	$@	$!	#%	#$	…

b)

Division	Quotient	Remainder
419/5	83	4 (!)
83/5	16	3 (@)
16/5	3	1 ($)
3/5	0	3 (@)

Final Answer = @$@! Is the message we should transmit

c) %### = $(0222)_5 = (0 \times 5^3) + (2 \times 5^2) + (2 \times 5^1) + (2 \times 5^0) = (62)_{10}$

The counter price is 62

Q5)

$(33 + 51) \times 70 = 6510$

$((3r^1 + 3r^0) + (5r^1 + 1r^0)) \times (7r^1 + 0r^0) = 6r^3 + 5r^2 + 1r^1 + 0r^0$

$(3r + 3 + 5r + 1) \times (7r) = 6r^3 + 5r^2 + 1r$

$56r^2 + 28r = 6r^3 + 5r^2 + 1r$

$6r^3 - 51r^2 - 27r = 0$

$\boxed{r = 9}$ $r = -1/2$ $r = 0$

↳ 9-Base System

Q6)

Directly To Binary (2)

Number	Remainder
689	
344	1
172	0
86	0
43	0
21	1
10	1
5	0
2	1
1	0
0	1

Binary: 1010110001

First To Hexadecimal (16)

Number	Remainder
689	
43	1
2	11
0	2

$(2B1)_{16}$

00101011 0001 (Binary)

Hexadecimal requires much less time to complete.

Q7)

a)

Dec	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
18-Base	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	G	H	10	11

b) $(AA)_{18} = (10 \times 18^1) + (10 \times 18^0) = (190)_{10}$

$(F2F)_{18} = (15 \times 18^2) + (2 \times 18^1) + (15 \times 18^0) = (4911)_{10}$

$4911*190 = (933090)_{10} = (8FHG6)_{18}$

c) Profit per burger = Selling cost – Cost to make

$(ABC)_{18} = (10 \times 18^2) + (11 \times 18^1) + (12 \times 18^0) = (3450)_{10}$

Profit per burger = $4911 – 3450 = (1461)_{10} = (493)_{18}$

$(F)_{18} = (15)_{10}$

$15*1461 = (21915)_{10} = (3DB9)_{18}$

Q8)

X1 => $(48414E)_{16}$
X2 => $(4B4841)_{16}$

a) $(48414E)_{16}$ - $(4B4841)_{16}$ using 16's Complement

16's complement of 4B4841 = B4B7BF

	4	8	4	1	4	E
+	B	4	B	7	B	F
	F	C	F	9	0	D

Here there is no carry, answer is - (16's complement of the sum obtained FCF90D)
16's complement of FCF90D = 0306F3
= $-(0306F3)_{16}$

b) $(4B4841)_{16}$ - $(48414E)_{16}$ using 15's Complement

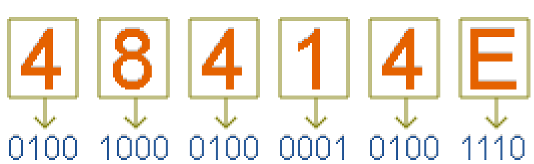
15's complement of 48414E = B7BEB1

		4	B	4	8	4	1
+		B	7	B	E	B	1
	1	0	3	0	6	F	2

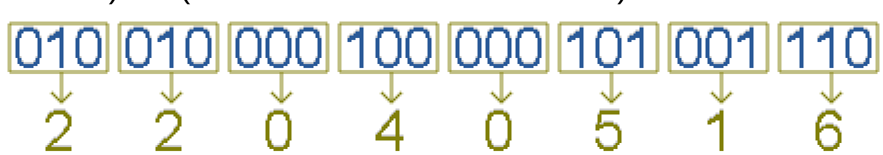
The left most bit of the result is called carry and add it to the rest part of the result 0306F2.
= 0306F2 + 1
= $(0306F3)_{16}$

c) Convert both X1 and X2 to radix 8 and find their sum

$(48414E)_{16}$ in octal:



$(48414E)_{16} = (010010000100000101001110)_2$



$(010010000100000101001110)_2 = (22040516)_8$
$(48414E)_{16} = (22040516)_8$

$(4B4841)_{16}$ in octal:

4	B	4	8	4	1
0100	1011	0100	1000	0100	0001

$(4B4841)_{16} = (010010110100100001000001)_2$

010	010	110	100	100	001	000	001
2	2	6	4	4	1	0	1

$(010010110100100001000001)_2 = (22644101)_8$

$(4B4841)_{16} = (22644101)_8$

Addition:

$(22040516)_8 + (22644101)_8$

$$\begin{array}{r} {\scriptstyle 1} \\ 22040516 \\ +\,22644101 \\ \hline 44704617 \end{array}$$

$= (44704617)_8$

Q9)

a) $450643_8 - 256752_8$

7's complement of 256752

$$
\begin{array}{r}
777777 \\
-256752 \\
\hline
521025
\end{array}
\xrightarrow{\text{8's complement of } 256752} 521026
$$

$$
\begin{array}{r}
1\quad\ \ 1\quad \\
450643 \\
+521026 \\
\hline
\cancel{1}171671
\end{array}
$$

↳ discarded overflow

$\boxed{171671}$ final answer.

b) $(1101.101)_2 - (1011.001)_2$

2's complement of 1011.001

$$
\begin{array}{r}
1111.111 \\
-1011.001 \\
\hline
0100.110 \\
+\qquad 1 \\
\hline
0100.111
\end{array}
$$

$$
\begin{array}{r}
1\ 1\ \ 1\ 1\ \ 1\ 1 \\
1101.101 \\
+0100.111 \\
\hline
\cancel{1}0010.100
\end{array}
$$

↳ discarded

$\boxed{0010.100}$ final answer.

Q10)

a) 123 + 456

$$\begin{array}{rcr} 123 & \rightarrow & 0001\ 0010\ 0011 \\ +456 & \rightarrow & +\ 0100\ 0101\ 0110 \\ \hline & & \underbrace{0101}_{5}\ \underbrace{0111}_{7}\ \underbrace{1001}_{9} \end{array}$$

$\boxed{579}$

b) 523 - 239

10's complement of 239 → 999 − 239 = 760 + 1

761

$$\begin{array}{rcr} 523 & \rightarrow & 0101\ 0010\ 0011 \\ 761 & & 0111\ 0110\ 0001 \\ \hline & & 1100\ 1000\ 0100 \end{array}$$

$$\begin{array}{r} 1100\ 1000\ 0100 \\ +\ 0110 \qquad\qquad \\ \hline 1\ \underbrace{0010}_{2}\ \underbrace{1000}_{8}\ \underbrace{0100}_{4} \end{array}$$

discard

$\boxed{284}$

c) 28 + 6.12

$$\begin{array}{rcr} 28 & & 0010\ 1000\ .\ 0000\ 0000 \\ +\ 6.12 & \rightarrow & +\ 0000\ 0110\ .\ 0001\ 0010 \\ \hline & & 0010\ 1110\ .\ 0001\ 0010 \\ & & +\ \quad 0110 \qquad\qquad\qquad \\ \hline & & \underbrace{0011}_{3}\ \underbrace{0100}_{4}\ .\ \underbrace{0001}_{1}\ \underbrace{0010}_{2} \end{array}$$

3 4 . 12

$\boxed{34.12}$

d) 0.127 + 1.99

$$\begin{array}{rcr} 0.127 & \rightarrow & 0000\ .\ 0001\ 0010\ 0111 \\ +\ 1.99 & & +\ 0001\ .\ 1001\ 1001\ 0000 \\ \hline & & 0001\ .\ 1010\ 1011\ 0111 \\ & & +\ \quad 0110\ 0110 \qquad \\ \hline & & \underbrace{0010}_{2}\ .\ \underbrace{0001}_{1}\ \underbrace{0001}_{1}\ \underbrace{0111}_{7} \end{array}$$

2 . 1 1 7

$\boxed{2.117}$

Q11)

a) $(-6)_{10} + (+5)_{10}$

Binary of 6 : 000110

Flip bits : 111001

2's Complement: 111010

$$\begin{array}{r} 111010 \\ +\ 000101 \\ \hline 111111 \end{array}$$

No carry bit answer is -ve

2's complement of answer : 000001 = $(-1)_{10}$

No overflow.

b) $(-1)_{10} - (+7)_{10}$

1 in 6-bit binary : 000001

2's complement : 111111

7 in 6-bit binary : 000111

2's complement : 111001

$$\begin{array}{r} 111111 \\ +\ 111001 \\ \hline 1\,111000 \end{array}$$

↳ discarded.

111000 2's complement 001000

↓

$\boxed{-8}$

No overflow.

c) $(+11)_{10} + (+20)_{10}$

11 in 6 bit 001011,

20 in 6 bit 010100

$$
\begin{array}{r}
001011 \\
+\ 010100 \\
\hline
011111
\end{array}
\rightarrow +(31)_{10}
$$

No overflow.

d) $(-23)_{10} + (-13)_{10}$

23 in 6 bit : 010111

2's complement : 101001

13 in 6 bit : 001101

2's complement : 110011

$$
\begin{array}{r}
1\ \ \ \ 11\ \ \\
101001 \\
+\ 110011 \\
\hline
1\ 011100
\end{array}
$$

carry bit

We have an overflow!

Q12)

Using a signed number the range is $(-2^{16-1}, 2^{16-1}-1)$ so the smallest binary integer is $(1000000000000000)_2$

In decimal the number is $(-32768)_{10}$

In Hexa Decimal the number is $(8000)_{16}$

In an unsigned number, the smallest would be $(0)_{16}$ which is $(0)_{10}$ in decimal and $(0)_{16}$

Q13)

Decimal	Signed Magnitude	Signed 1's Complement	Signed 2's Complement
-10	1000 1010	1111 0101	1111 0110
-128	*	*	1000 0000
-32	1010 0000	1101 1111	1110 0000
+64	0100 000	0100 000	0100 000
+127	0111 1111	0111 1111	0111 1111
+0	0000 0000	0000 0000	0000 0000

Q14)

Find out the even parity bit of the following binary numbers:

I. 01010111

Number of 1's = 5 (Odd) => Even Parity bit is 1

II. 10110100

Number of 1's = 4 (Even) => Even Parity Bit is 0

Q15)

a) Add 3 or 0011 to each to 4 digit BCD Code

0110100101110011 -> **Excess-3: 1001110010100110**

b) MSB or left most digit is the same, then proceed to XOR from current bit and next bit to obtain next grey code bit (from left to right or from MSB to LSB)

0100100111110101 -> **Grey Code: 0110110100001111**